Vinay Premchandran Nair

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EDUCATION

Carnegie Mellon University (CMU) **Pittsburgh, U.S.**

Master of Science in Artificial Intelligence and Innovation (MSAII) *Aug 2022 - May 2024*

●Coding Bootcamp, Distributed Systems, AI Engineering, ML in Production, Deep Learning, Machine Learning, NLP, Multimodal ML

Pune Institute of Computer Technology (PICT) **Pune, India**

Bachelor of Engineering in Information Technology *Aug 2018 - July 2022*

●Data Structures & Algorithms, Object Oriented Programming, Database Management, Cloud Computing, Computer Architecture

RELEVANT WORK EXPERIENCE

Senior Data Scientist | Apple, Inc. U.S.A. (V) | Fractal.ai *June 2025 - Present*

● Architected a **LangGraph**-based multi-agent system that cut Apple's analysts' insight turnaround from ~2 weeks to under 10 minutes, automating the end-to-end querying of **apple.com** engagement metrics across millions of datapoints .

● Engineered dual-vector retrieval pipelines:

→ **Structured store indexing** of aggregate Snowflake tables (sampling **~10000** rows per query, column-description embeddings).

→ **Unstructured store indexing** >**80** Keynote slides chunked semantically.

● Built an adaptive "**experience**" planner agent performing dual-vector lookups to route tasks between structured and unstructured tools; orchestrated in LangGraph with persistent Redis context caching (migrated from Memcache, enabling TTL-based session recall and **2× faster context fetch**).

Machine Learning Engineer (Capstone Project) | Bank of New York Mellon, U.S.A *Jan 2024 - May 2024*

● Developed BNY Mellon's **Knowledge Graph** based **RAG** using **Neo4J** and **QDrant**, achieving a **10%** improvement in **reasoned retrieval** with novel **Contriver** embeddings augmenting the graph over regular vector databases using **LlamaIndex**

● Enhanced Financial **Multihop QA** accuracy by **2%** through Embeddings Augmented Knowledge Graph

● Boosted accuracy by **20%** in automated financial responses through **Microsoft Autogen** for **multi-agent LLM interactions**

Data Scientist | C3.ai, U.S.A. *Jan 2025 - June 2025*

● Spearheaded a hybrid **RAG** pipeline in **PostgreSQL** using the **PGVector** plugin, leveraging **GIN indexing** for keyword filtering, **B-Tree** for metadata filtering, and **IVFFlat** indexing for vector search, boosting retrieval precision by **40%**.

● Built a user-configurable reranker (cross-encoder & LLM-based), improving passage ranking by **25%** across filtered retrievals.

● Built a Root Cause Analysis tool for RAG failures, saving **45–50 mins/query** of debugging and reducing manual effort by **80%**.

Machine Learning Engineer | Capgemini, U.S.A. *July 2024 - Jan 2025*

● Enhanced legal document metadata extraction by **30%** using **AWS Textract** and **Claude 3.5**, processing over **1000** documents monthly

● Streamlined metadata storage and retrieval with AWS Lambda and **DynamoDB**, reducing process time by **25%**

● Developed a **ReactJS** web application with **AWS Lambda** integration, supporting daily operations for over **100** users

● Enhanced system monitoring with AWS CloudWatch and Splunk, managing 3,000+ operational events daily for optimal performance

RELEVANT PROJECTS

Tip Of the Tongue (ToT) Retrieval using LLMs (Advisors: Dr. Chenyan Xiong and Dr. Daphne Ippolito)

● Designed a **GPT-4** based **Query Decomposition** framework that improved query performance for movie search engines

● Implemented a **Dense Retrieval Model** to increase search efficiency and accuracy in matching movie names

● Developed a **Pointwise Reranking Mechanism** with GPT-4, independently scoring retrieved results for optimized ranking

● Improved key metrics, raising NDCG from **10% to 23%** and Precision from **4% to 20%** in search results

Augmenting Multimodal Multihop Question Answering (Advisor: Dr. Louis-Philippe Morency)

● Implemented a patch-focused **Vision-Language Transformer** and **T5** answer generator, resulting in an **8%** QA accuracy gain

● Implemented a **hierarchical approach for dense data representation**, improving multimodal reasoning capabilities by **2%**

● Engineered **MiniGPT-4** with Vision Transformer and Llama-V2 for advanced multimodal **Chain of Thought Reasoning**